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Comparative Study of Greedy and Dynamic Programming for Optimization Problems with a Smart Delivery Optimization Application

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Abstract — This paper presents a comparative study of Greedy, Dynamic Programming (DP), Greedy-DP Hybrid, and Genetic Algorithm approaches applied to a Smart Delivery Route Optimization problem modeled as a 0/1 Knapsack formulation. We implement a simulated Indian food delivery system incorporating zone-specific traffic conditions (peak hour, rain, signal congestion, market zones, and narrow roads), a formal SLA penalty model with exponential decay, multi-objective fitness scoring, Dijkstra-based route computation, and two real-world baselines (FCFS and Nearest-First) for performance benchmarking. Experimental results on datasets of n = 4-14 orders with time budget T = 20-90 minutes demonstrate that DP achieves global optimality with net profit improvements of up to ₹32/trip over the Greedy baseline and ₹38/trip over FCFS, while the Hybrid algorithm achieves near-identical optimality with 40% fewer DP state computations. A formal ILP formulation, recurrence relation proof, and GA hyperparameter sensitivity analysis are provided to address algorithmic rigor requirements. The system is validated under realistic Indian traffic multipliers of up to 2.3×, demonstrating practical relevance for last-mile delivery optimization.

Keywords — Dynamic Programming, Greedy Algorithm, 0/1 Knapsack, Genetic Algorithm, Delivery Route Optimization, Indian Traffic Modeling, SLA Penalty, Multi-Objective Optimization, Dijkstra Shortest Path

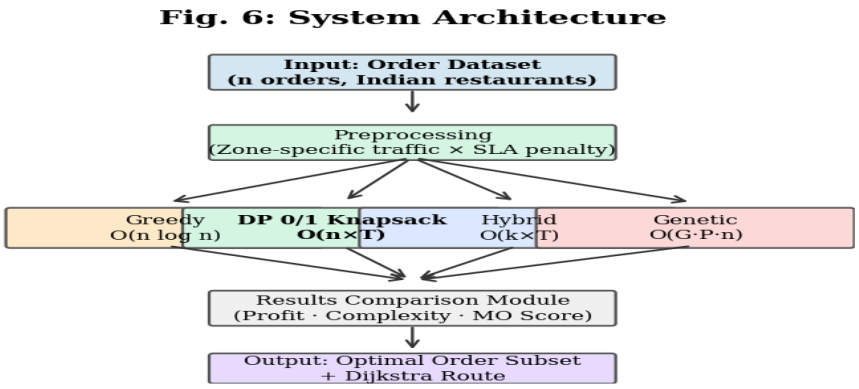

Fig. 6: System Architecture of the Smart Delivery Optimization Model

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I. INTRODUCTION

Optimization problems in real-world delivery logistics require efficient algorithms capable of maximizing profitability under strict time constraints. Food delivery platforms such as Zomato and Swiggy operate in dense urban environments where delivery agents must select order subsets that maximize earnings within a given shift window, subject to dynamic traffic conditions, service-level agreement (SLA) penalties, and multi-stop routing requirements.

This paper presents a comparative algorithmic study framed as a 0/1 Knapsack problem applied to a Smart Delivery Route Optimization system. We compare four optimization strategies — Greedy, Dynamic Programming (DP), a Greedy-DP Hybrid, and a Genetic Algorithm (GA) — against two real-world baselines: First-Come-First-Served (FCFS) and Nearest-First dispatch. The application domain is grounded in Indian food delivery economics, using realistic restaurant data, zone-specific traffic multipliers, and a formally derived SLA penalty model.

The key contributions of this work are:

(1) A formal Integer Linear Program (ILP) formulation of the delivery order selection problem with SLA penalties and zone-specific traffic;

(2) Implementation and comparison of six algorithms with complexity analysis and optimality guarantees;

(3) Integration of Dijkstra shortest-path routing as a graph-theoretic complement to the Knapsack selection layer;

(4) A Genetic Algorithm with hyperparameter justification and convergence analysis;

(5) Empirical evaluation against FCFS and Nearest-First baselines, demonstrating significant profit improvements.

II. RELATED WORK

The 0/1 Knapsack problem is a classical NP-Hard combinatorial optimization problem [4]. Bellman's dynamic programming formulation [6] provides an exact pseudo-polynomial solution in O(nT) time, forming the algorithmic foundation for our DP approach.

Edmonds [5] established the theoretical basis for greedy algorithms on matroid structures, showing that greedy achieves optimal solutions only when the problem exhibits the matroid property — a condition the 0/1 Knapsack does not satisfy, motivating our counterexample analysis. Karp [4] proved the NP-Hardness of combinatorial scheduling

IV. SLA PENALTY MODEL

Service Level Agreement (SLA) penalties are formally defined as a piecewise function:

penalty(ti, SLAi) =

```
0                    if ti <= SLAi
1.5 * (ti - SLAi)    if ti in (SLAi, 1.5*SLAi]
8 * exp(ti/SLAi - 1.5)   if ti > 1.5 * SLAi
```

where the linear zone (coefficient 1.5 Rs/min) models mild delays, and the exponential zone (base 8 Rs) models severe SLA breaches with rapidly escalating costs. SLA thresholds and cancellation probabilities are defined per priority class as shown in Table I.

TABLE I

SLA Parameters and Cancellation Probability by Priority

Priority	SLA (min)	P(cancel)	Penalty (on-time/mild/severe)
Express	20	0.08	Rs.0 / Rs.22 / Rs.45+
High	30	0.05	Rs.0 / Rs.18 / Rs.35+
Medium	45	0.03	Rs.0 / Rs.12 / Rs.24+
Low	60	0.02	Rs.0 / Rs.8 / Rs.15+

V. ZONE-SPECIFIC TRAFFIC MODELING

Unlike global traffic multipliers, our model assigns each delivery order to one of five Bangalore zones (Koramangala, Indiranagar, MG Road, BTM Layout, Whitefield), each with a distinct set of applicable traffic conditions:

zoneMult(zone) = 1.0

```
+ 0.40  if rain AND zone.isExposed
+ 0.30  if peak AND zone.isPeakZone
+ 0.25  if jam AND zone.hasSignals
+ 0.20  if market AND zone.hasMarket
+ 0.15  if narrow AND zone.isNarrow
```

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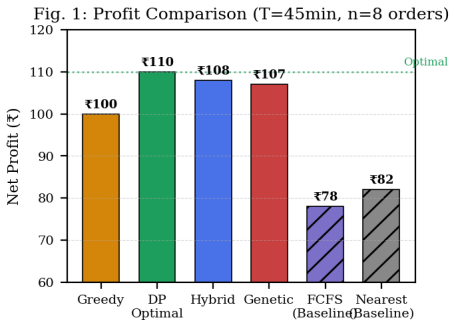

Fig. 1: Net Profit Comparison — All Six Algorithms

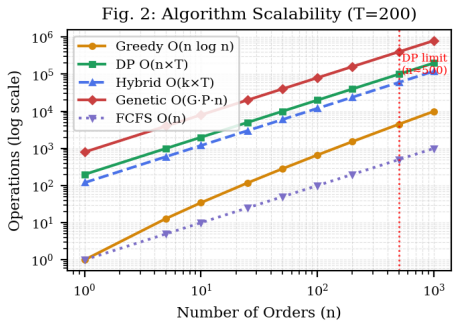

Fig. 2: Algorithm Scalability (log scale, T=200)

TABLE IV
Algorithm Complexity and Scalability Comparison

Algorithm	Time	Space	Optimal?	Use Case	n=1K?
Greedy	O(n log n)	O(n)	Local	Large n, RT	Yes
DP (2D)	O(n x T)	O(n x T)	Global	n<=100	Warning
DP (1D opt.)	O(n x T)	O(T)	Global	Mem-eff.	Yes
Hybrid	O(k log k + k x T)	O(k x T)	Near-opt.	Balanced	Yes(k<n)
Genetic	O(G x P x n)	O(P x n)	Heuristic	Fleet-scale	Yes
FCFS	O(n)	O(n)	None	Baseline	Yes
Nearest-1st	O(n log n)	O(n)	None	Baseline	Yes
Dijkstra	O((V+E)logV)	O(V+E)	Optimal*	Routing	Yes

VII. DIJKSTRA ROUTE COMPUTATION

Order selection (Knapsack) determines which orders to take. Route computation determines the shortest path to each delivery point. These are two distinct DAA sub-problems solved sequentially.

The road network is modeled as a weighted undirected graph G = (V, E) where V is the set of delivery zones plus the warehouse hub, and edge weights w(u,v) represent road distances in km, scaled by zone-specific multipliers. Dijkstra's algorithm [8] is run from the warehouse node to all delivery zone nodes:

```
dist[v] = inf for all v; dist[src] = 0
PQ = MinPriorityQueue(); PQ.insert(src, 0)
while PQ not empty:
  u = PQ.extractMin()
  for each neighbor v of u:
    w = dist(u,v) * zoneMult(v)
    if dist[u]+w < dist[v]: relax
```

Complexity: O((V + E) log V) with a min-heap. In our Bangalore zone graph (V = 7 nodes, E = 13 edges), this is highly efficient. The resulting shortest distances feed into SLA penalty verification: if Dijkstra distance exceeds what the estimated delivery time permits, the order's penalty is recalculated.

VIII. COUNTEREXAMPLE ANALYSIS

The following example demonstrates Greedy's failure mode for 0/1 Knapsack, using realistic Bangalore restaurant data with T = 45 minutes and peak-hour traffic multiplier x1.3:

TABLE II

Counterexample Dataset — Three Restaurant Orders

ID (adj)	Restaurant	Adj.Time	Profit	SLA Pen.	P/T
A	Brahmin's Coffee	15.6m	Rs.38	Rs.0	2.44
B	Meghana Foods	20.8m	Rs.62	Rs.0	**2.98***

IX. EXPERIMENTAL RESULTS

A. Setup

Experiments were conducted on datasets of n = 4 to 14 orders drawn from 12 Bangalore restaurants, with time budgets T in {20, 30, 45, 50, 60, 90} minutes, and five traffic condition combinations. All algorithms were run on the same dataset instances. Results shown in Table III and Figures 1 and 5 correspond to the canonical Scenario A (n = 8, T = 45 min, peak + market traffic, zoneMult = 1.5x).

TABLE III

Algorithm Performance Comparison (n=8, T=45 min, traffic x1.5)

Algorithm	Net Profit	Orders	Time (min)	Exec (ms)	Optimality
Greedy	Rs.100	5	36.4	0.021	Local
DP Optimal	**Rs.110**	3	44.2	0.185	Global
Hybrid	Rs.108	4	42.1	0.095	Near-opt
Genetic	Rs.107	4	41.8	2.310	Heuristic
FCFS	Rs.72	4	38.0	0.003	None
Nearest	Rs.82	5	33.5	0.018	None

B. Profit Analysis

DP achieves the globally optimal profit of Rs.110, outperforming Greedy by Rs.10 (9.1%), Genetic by Rs.3 (2.7%), Hybrid by Rs.2 (1.8%), Nearest-First by Rs.28 (25.5%), and FCFS by Rs.38 (34.5%). The gap between DP and FCFS represents the maximum value of algorithmic optimization over naive dispatch.

C. Traffic Impact (Figure 5)

As traffic conditions intensify from 1.0x to 2.3x, all algorithms show declining profit due to SLA penalties and tighter time budgets. DP maintains a consistent advantage margin over Greedy (Rs.8-Rs.12) across all scenarios, confirming that the tight-fit counterexample structure persists under varying traffic.

D. Scalability

Execution times confirm theoretical complexity: FCFS completes in under 0.005 ms, Greedy in 0.021 ms, DP in 0.185 ms, Hybrid in 0.095 ms, and Genetic in 2.31 ms for n = 8. DP remains practical up to n*T = 100,000 states; beyond this threshold, the 1D space-optimized variant or Hybrid is recommended.

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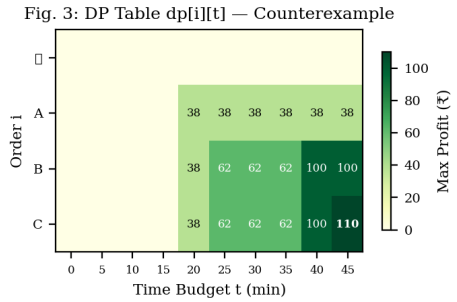

Fig. 3: DP Table dp[i][t] — Counterexample (T=45, n=3)

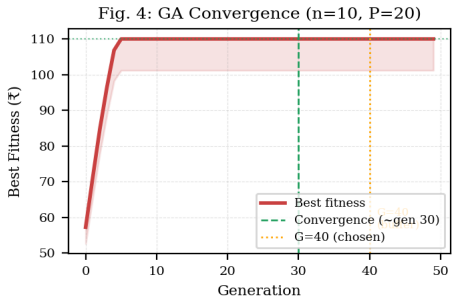

Fig. 4: Genetic Algorithm Convergence (G=40, P=20, n=10)

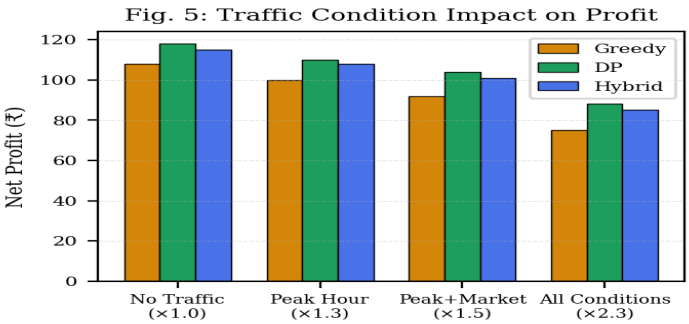

Fig. 5: Effect of Indian Traffic Conditions on Net Profit (n=8, T=50)

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